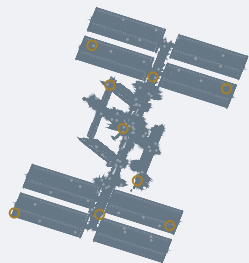


Viability-preserving rollout ADP for orbital inspection

a Mission inputs



○ Required targets K



Camera model

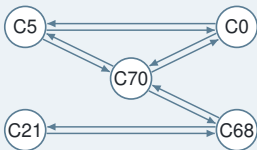
Initial planning state

$$s = (j, m, b, h)$$

Start pose x_0 ; budget H

b Audited SOOA graph

Viewpoints + visibility masks



Directed HCW transfers

Dynamics · clearance · arrival

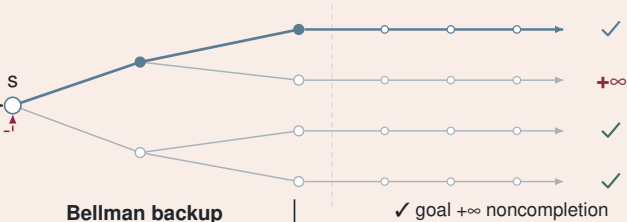
✓ Audit passed

Arc cost ℓ ; admissibility χ

c Rollout ADP

Exact prefix: depth d

Complete base-policy tails



Update state and replan

d Replanning

First action a^*

Minimum finite value

Append arc + update

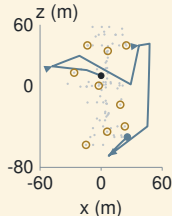
$$s' = f(s, a^*)$$

Required goal reached?

Yes

No

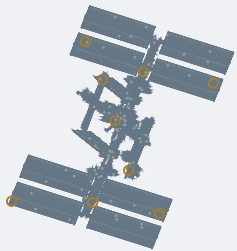
Planned route



Required targets: 9/9

Orbital inspection: from required targets to a complete plan

a Mission



○ **Required targets K**

Start pose x_0 ; budget H

b Construction



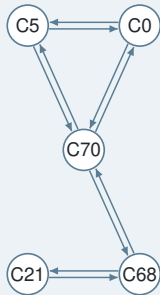
Visibility masks

HCW transfers



Motion audits

c Graph



Audited actions

d Rollout ADP

Lookahead d

Complete policy tails

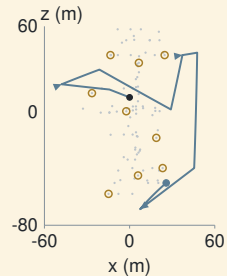


✓ goal $+\infty$ noncompletion

State update + replan

Min-cost first action

e Plan



Required targets: 9/9

Viability-preserving rollout ADP: decision and state update

a Planning state

$$s = (j, m, b, h)$$

j current viewpoint
m covered targets
b selected views
h remaining budget

Goal: cover K
 $\Gamma_K(m) = 1$

b Safe actions

Audited + unvisited



$$\mathcal{U}_s(s): b_a = 0, \chi_{ja} = 1$$

--- X Unsafe
— X Visited

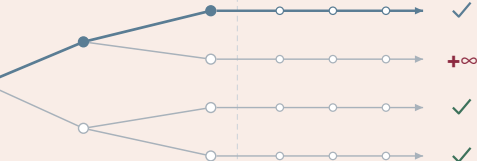
**Safe connecting
actions retained**

c Rollout value evaluation

Exact prefix
depth d

Complete base policy μ
to goal or budget limit

s



Bellman backup

✓ goal +∞ noncompletion

$$\hat{Q}_d(s, a) = \ell_{ja} + \hat{V}_{d-1}(f(s, a))$$

d Policy update

Finite values only

$$a^* = \arg \min_a \hat{Q}_d(s, a)$$

First action a^*

**Append selected arc
and update state**

$$s' = f(s, a^*)$$

$$h \leftarrow h - 1$$

All values +∞
No certified completion

Complete plan

Yes

**Required goal
reached?**

No: update and replan